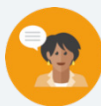


1.6 Transformations on the Plane – Part 2

Lesson Introduction

	Benchmark	MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.			Learning Target	I can describe a single transformation that was applied.
	Benchmark Coherence	Building On MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.		Working Toward N/A		
	Essential Questions	- What are differences between the transformations? - How can each transformation be identified? - What is the simplest way to describe each transformation?				
	Lesson Narrative	This lesson is a culmination of all the transformations in the previous lessons. Students progress through the lesson by identifying and describing the transformation that results in an image from a given preimage. Students are led to being able to identify and describe all transformations.				
	Component Summary	1.6.1 Warm-Up (~5 min.)	Which polygon does not belong? (SE p. 34)	1.6.3 Guided Practice (5 min.)	Describing a transformation (SE p. 37)	
		1.6.2 Guided Instruction (25 min.)	Identifying and describing single transformations (SE p. 35)	1.6.4 Your Turn! (5 min.)	Describing a transformation (SE p. 39)	
		1.6.5 Wrap-Up (~10 min.)	Identifying and describing single transformations (SE p. 40)			
	Resources	Glossary		Materials		Additional Preparation
		<u>Key Terms:</u> - N/A <u>Additional Terms:</u> - Translation - Rotation - Reflection - Dilation		(Optional) Patty Paper (Optional) Ruler		N/A

 Teacher Tips	Common Misconceptions Students may believe that all transformations are rigid transformations.	Things to Consider - If students are struggling with finding the point of dilation on 1.6.2 Guided Instruction question 4, consider starting with images where the point of dilation is the center of the image. Once this is something students can understand, consider moving it to a point on the edge of the image. Finally, try one where the point is outside of the image. - If time is an issue, then consider assigning 1.6.4 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Number Talk For 1.6.2 Guided Instruction, display each transformation one at a time. Give the students a few minutes to quietly think and give a signal when they have an answer. Select students to share their strategies for how they approached the problem. Press students to provide details they see that have not already been shared.	MA.K12.MTR.1.1: Participate in Effortful Learning Identifying transformations can be a difficult and challenging task. Students will build perseverance by listening to others and modifying their methods for determining a transformation. Having students share their methods will encourage engagement and allow others to make sense of the task.
 Differentiate Instruction	The transformations that are present throughout this lesson starting at 1.6.2 Guided Instruction could be represented with physical objects. This could be patty paper, Tangrams, Play-Doh, or other manipulatives.	
	In 1.6.4 Your Turn, encourage students to discuss alternative methods of getting to the same answer. This can lead pairs or groups to discuss the topic and come up with alternatives that otherwise may not have been thought of.	
Lesson Notes:		

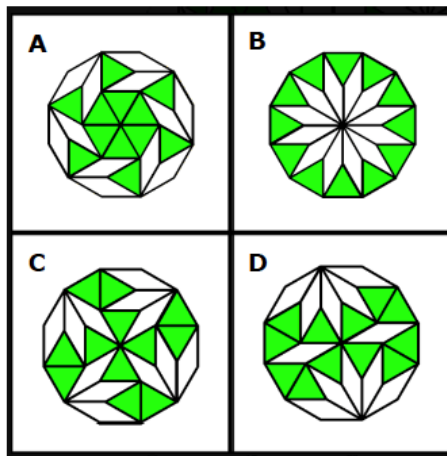
1.6.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students identify which polygon they believe does not belong. Each polygon has a reason why it does not belong. The selection students make is not important, but the reason why is what should be highlighted.



1.6.1 Warm-Up: Which One Doesn't Belong?

- The diagram shows four polygons.



- Which polygon does not belong?

Answers will vary.

- Describe your reasoning.

Answers will vary. Polygon A has a perfect hexagon. Polygon B has all white spaces sharing a line. Polygon C has no more than three shapes sharing a side. Polygon D looks as if it has a mirror in the middle.



There is no right or wrong answer. Help students understand that the purpose is for them to consider which does not belong to defend their choice. Encourage students to have flexible thinking.



Students could be challenged to stretching their reasoning to explain why another option could also be the answer. This would allow students to consider another perspective that they may have not seen before.

1.6.2 Guided Instruction: Determining a Single Transformation

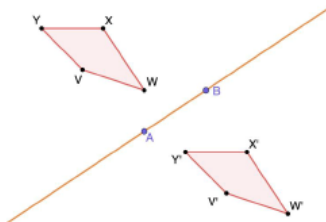
Component Overview: Given a preimage and an image, students identify and describe the type of transformation used. Students will complete a chart of how they distinguish between the four transformations.



1.6.2 Guided Instruction: Determining a Single Transformation

1. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.

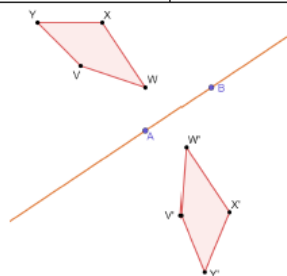
Describe the single transformation that was used to create $V'W'X'Y'$.



Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☒ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☒ Down ☐ Left ☒ Right ☐ Up	

2. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.

Describe the single transformation that was used to create $V'W'X'Y'$.



Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☒ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☒ Line of reflection is $\overline{AB}$. ☐ Down ☐ Left ☐ Right ☐ Up	



Students may benefit from having a piece of patty paper to help them determine the transformation. For many students, visualization of transformations is difficult and should be supported with tools, such as patty paper.



If students choose the incorrect transformation, consider asking the following question.

- What do you notice about the distance between each point of the preimage to its corresponding point in the image?
- If this were a reflection or rotation, where would vertex W be in relation to the line segment or the points?

If students are correct, asking these questions will help them defend their reasoning.



If students choose the incorrect transformation, consider asking the following question.

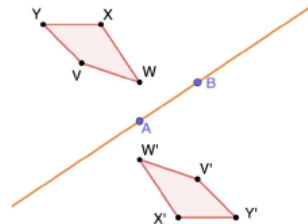
- What do you notice about the orientation of the vertex points in the preimage compared to the order of the vertex points in the image?

If students are correct, asking this question will help them defend their reasoning.

1.6.2 Guided Instruction: Determining a Single Transformation (continued)

3. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.

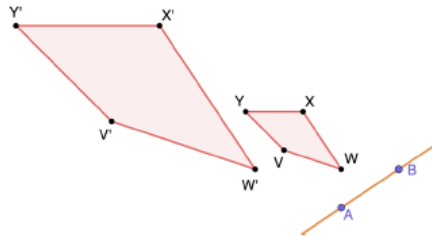
Describe the single transformation that was used to create $V'W'X'Y'$.



Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☒ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☒ Center of rotation is A . ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	180° rotation clockwise or counterclockwise

4. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.

Describe the single transformation that was used to create $V'W'X'Y'$.



Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☒ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☒ Center of dilation is B . ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	Scale factor of 2.5



If students are having a hard time finding the point of rotation, students can draw line segments connecting each point of the preimage to its corresponding point on the image. Where the four lines intersect is the point of rotation.

To verify a point of rotation, students can place the point of a compass on a possible “point of rotation.” If the point is correct, the compass can be opened to any specific point of the preimage and an arc drawn that will perfectly intersect with the corresponding point on the image.



Students may need to be reminded that rotation can be clockwise or counterclockwise.



To find the point of dilation, students should draw lines extending through each point of the preimage and the corresponding point on the image. The point of dilation is where these four lines intersect.

Students can determine the scale factor by finding the actual measurements or estimating using patty paper.

1.6.2 Guided Instruction: Determining a Single Transformation (continued)

5. Complete the chart so it can be used to distinguish between similar-looking transformations. First, write your own hints. Then, exchange hints with your partner.

Transformation	My Hints	My Partner's Hints
Dilation	<i>Answers will vary.</i>	<i>Answers will vary.</i>
Reflection	<i>Answers will vary.</i>	<i>Answers will vary.</i>
Rotation	<i>Answers will vary.</i>	<i>Answers will vary.</i>
Translation	<i>Answers will vary.</i>	<i>Answers will vary.</i>



When monitoring student responses to question 5, make note of students who have different hints. Call on these students to share their hints with the whole class. Students benefit from hearing the thinking of others.

6. Which transformations resulted in congruent figures?

- ☐ Dilation
- ☒ Reflection
- ☒ Rotation
- ☒ Translation

7. When a transformation results in a congruent figure, what is another name that can be used for that transformation?

The transformation that results in a congruent figure is a rigid transformation.



Another name for a transformation that results in a congruent figure is an isometry.

1.6.3 Guided Practice: Describing Transformations

Component Overview: Students identify and describe each transformation that results in an image from a given preimage.

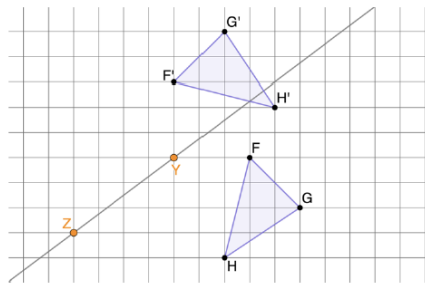


1.6.3 Guided Practice: Describing Transformations

Each of the diagrams shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.

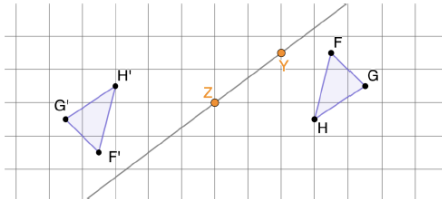
- Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.

Rotate 90° counterclockwise around point Y or Rotate 270° clockwise around point Y



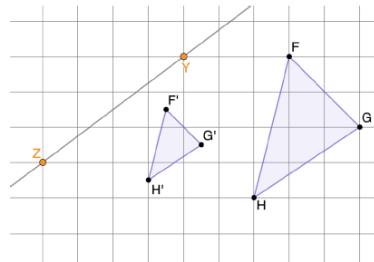
- Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.

Rotate 180° counterclockwise around point Z.



- Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.

Dilation with scale factor of $\frac{1}{2}$ and center of dilation point Z.



- Review your hints chart and add anything else you think will be helpful.

Answers will vary.



Allow informal observation throughout the 1.6.2 Guided Instruction to dictate how “guided” 1.6.3 Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



Consider challenging students to find another way to end up with the same post-image without using rotation.



How can finding the slope between the center of rotation and a point on the preimage be compared to the slope between the center of rotation and the corresponding point on the image to find the scale factor?

1.6.4 Your Turn!

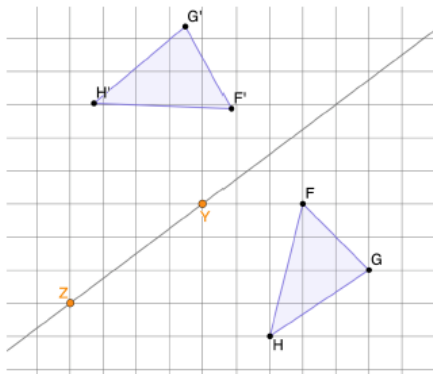
Component Overview: Students identify and describe each transformation that results in an image from a given preimage.



1.6.4 Your Turn!

1. The diagram shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.

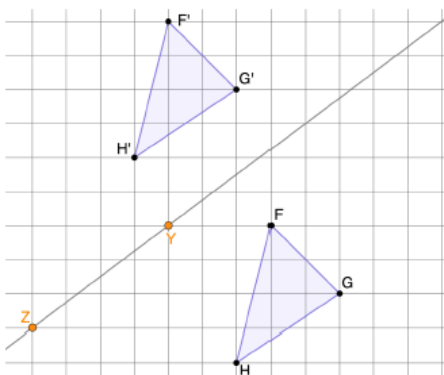
Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.



The transformation that was applied was a reflection over line $\overleftrightarrow{YZ}$.

2. The diagram shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.

Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.



The transformation was a translation of six units up and three units left.



Monitor while students are working to determine if there are any questions that need to be debriefed before moving to this component. This could include highlighting student voices to share rich discussions or comments overhead while you circulated or to address common misconceptions that were evident among the majority of the class.

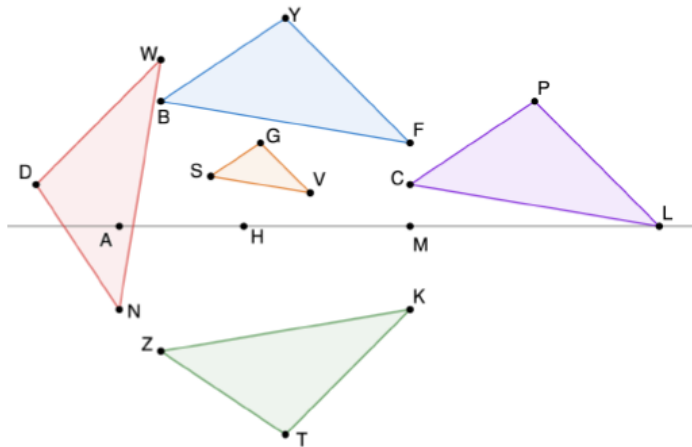
1.6.5 Wrap-Up: Ticket Out the Door

Component Overview: Students summarize their learning by identifying and describing the transformation that will produce each image.



1.6.5 Wrap-Up: Ticket Out the Door

- The diagram shows four different single transformations of $\triangle YFB$.



Complete the chart by naming and describing the type of transformation that was applied to $\triangle YFB$ to result in each triangle.

Triangle	Type of Transformation	Description of the Transformation
$\triangle PLC$	Translation	The transformation was a slide down and shifted to the right.
$\triangle TKZ$	Reflection	It was a reflection over line $\overline{HM}$.
$\triangle GVS$	Dilation	The triangle has a center of dilation at point H.
$\triangle DWN$	Rotation	The rotation is 90° counterclockwise at point H.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.



Be careful to check each type of transformation along with the description. Students may see other paths to the correct answer.